L A I B A K H A W A R

AI Engineer | LLM & NLP Systems | Backend & Data Engineering
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PROFESSIONAL SUMMARY

AI Engineer building LLM applications, autonomous agents, graph-based retrieval, and scalable Python services. Delivers evidence-backed outputs and reliable workflows using FastAPI, PostgreSQL, and Docker, with experience in model evaluation, data pipelines, source-code auditing, and technical handover.

TECHNICAL SKILLS

AI & NLP: LLM agents, RAG, embeddings, semantic search, fine-tuning, summarization, NLI, LLM-as-a-judge, multilingual NLP; LangChain, Hugging Face, PyTorch, Sentence Transformers, FAISS, Ollama, Scikit-learn

Backend & Data: Python, FastAPI, Flask, REST APIs, Pydantic, SQL, Pandas, NumPy, ETL, Spark, Redis, Celery; PostgreSQL, pgvector, SQL Server, Neo4j, Memgraph, MySQL, MongoDB

Cloud & Delivery: Azure AI, AWS (EC2, S3, Lambda, Glue, Redshift, API Gateway), Docker, Git, GitHub, Streamlit, Next.js, React, TypeScript, Vapi; source-code review, QA, technical documentation

PROFESSIONAL EXPERIENCE

AI Engineer | SysReforms International

Feb 2026 - Present | *Islamabad, Pakistan*

- Develop PVS-LLM pipelines for summarization, question answering, concept extraction, and semantic mapping; build multilingual FastAPI/Streamlit applications using Mistral, Azure AI, PostgreSQL, and Docker.
- Build evaluation using semantic similarity, NLI, BERT-based metrics, and LLM-as-a-judge; improve execution through batching, parallel processing, and resumable workflows.
- Complete 719 V9 review requirements and 65 V12 corrections across 4 system areas; create source-traceable documentation and audit registers, review 392 PDF pages, and audit database operations and security risks.

AI Backend Engineer | Dataspecc

Jan 2025 - Jan 2026 | *Islamabad, Pakistan*

- Built Hyperengage's Customer Memory Graph with Neo4j/Memgraph, modelling accounts, contacts, commitments, risks, deals, and time-bounded relationships.
- Combined pgvector retrieval with graph traversal and LLM-generated account memory; integrated HubSpot, Salesforce, Segment, Intercom, and Zendesk data; contributed to Signals Engine and Copilot Q&A.

Teaching Assistant - NLP and Database Systems | FAST NUCES

Feb 2025 - Jun 2025 | *Islamabad, Pakistan*

- Supported undergraduate lectures, lab sessions, assignments, and assessment in NLP and Database Systems.

Machine Learning Intern | Devsort

Jun 2024 - Aug 2024 | *Islamabad, Pakistan*

- Supported data collection, preprocessing, exploratory analysis, model development, and testing using Python, Pandas, NumPy, Scikit-learn, and Matplotlib.

FEATURED PROJECTS

AnalystOS Autonomous Data Analyst - Built an agent with 18 typed tools for CSV/Excel and SQL analysis, charts, and evidence-backed reports; added read-only SQL validation, isolated Python execution, anomaly detection, and numeric-claim verification. GitHub: AnalystOS

Veriflow AI Investigation & Agent Orchestration - Built a 7-agent risk-investigation platform demonstrated on 3,000 synthetic accounts, with persistent task DAGs, retries, approval gates, and 3-stage claim verification. GitHub: AgenticAIPlatform

Voice AI Patient Registration - Built phone-based intake with Vapi, GPT-4.1, FastAPI, PostgreSQL, and Next.js; enforced validation, caller confirmation, duplicate detection, and idempotent saves. GitHub: voice-ai-patient-registration

AI Engineering Portfolio - Built browser-based ML demos, project case studies, and a retrieval lab combining BM25, MiniLM embeddings, and graph expansion. www.laibakhawar.com

Hybrid Rule-Based & LLM-Assisted EDI Mapping - Combined deterministic rules, LLaMA 3.1 assistance, and syntax/consistency validation for Tradacoms, X12, and HL7 conversion.

Conversational RAG & Research Paper Agent - Built document QA using FAISS, embeddings, reranking, query rewriting, conversation memory, PDF parsing, summarization, and tool calling.

EDUCATION & CERTIFICATIONS

BS Data Science | FAST NUCES, Islamabad | 2021 - 2025 | Dean's List: Spring 2023, Fall 2024

A Levels in Engineering | Beaconhouse Margalla Campus, Islamabad

Certifications: LangChain: Chat with Your Data | AWS Web Application Builder | AWS Data Pipeline Builder | DataCamp: Understanding Data Visualization | DataCamp: Introduction to Data Science in Python